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Reality Vector:

A Universal 10-Dimensional Environmental State Model

for Autonomous Systems.

One Representation to Rule All Physical Domains.

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Abstract

Autonomous systems operating in physical environments must maintain a representation of environmental state that is complete enough to support safe decision-making. Existing state models are application-specific and incomparable across domains: a robot's state model and an agricultural system's state model share no common structure. This paper presents the **Reality Vector**: a universal 10-dimensional environmental state model applicable across all physical domains. Each dimension represents a fundamental property of physical reality — spatial position, temporal state, material composition, energy state, informational state, causal state, boundary conditions, uncertainty, agency, and outcome — and the vector's 10 components are defined formally and independently of application domain. We prove that the Reality Vector is universal: any physically meaningful environmental state can be represented as a Reality Vector, and any decision that is safe given a Reality Vector representation is safe given the actual environmental state. We present the formal definitions of all 10 dimensions, the projection operators for domain-specific state extraction, and deployment evidence from an agricultural autonomous system showing a 23% improvement in decision quality and a 40% reduction in state-related software bugs.

This paper is a technical specification and position paper. Empirical validation data is drawn from the Agentic platform's production deployment and is reported as operational statistics subject to standard system monitoring limitations.

Figure 1: The Reality Vector $\vec{R} = (S, T, M, E, I, C, B, U, A, O)$: a universal 10-dimensional environmental state representation. Each dimension captures a fundamental property of physical reality independent of application domain. The vector is defined formally and the projection operators $\pi_d(\vec{R})$ extract domain-specific state slices.

Dim	**Name**	**Records**
S	Spatial Position	Location vectors, orientations, uncertainty ellipsoids for all tracked entities
T	Temporal State	System time, epoch of last update, time horizon of current estimate
M	Material Composition	Density, stiffness, conductivity, porosity per entity
E	Energy State	Stored energy per entity, energy flux across boundaries, energy constraints
I	Informational State	Knowledge flags (sensed vs. believed) per entity-attribute pair
C	Causal State	Causal relationships: which entities/events influence which others
B	Boundary Conditions	Physical laws, hard constraints, resource limits in force
U	Uncertainty	Entropy distribution across all other dimensions
A	Agency	Entities capable of autonomous action, available actions per entity
O	Outcome	Expected outcome trajectory, utility function governing preference

Keywords: reality vector, environmental state model, autonomous systems, spatial reasoning, universal representation, agricultural AI, multi-domain AI, BX3 Framework, dimensional analysis

1 Introduction

An autonomous system operating in a physical environment must answer the question: *what is the actual state of the world?* The answer requires a representation — a data structure that captures the relevant properties of the environment in a form the system can reason about. The quality of this representation determines the quality of the decisions the system can make.

Existing representations are application-specific. A robot's state model contains pose, velocity, joint angles, and end-effector position. An agricultural autonomous system's state model contains soil moisture, crop stage, weather conditions, and irrigation status. A medical autonomous system's state model contains patient vital signs, drug concentrations, and physiological parameters. These representations cannot be compared or composed across domains: there is no common structure on which a unified decision-making architecture can be built.

This domain-specificity creates a fundamental interoperability problem. When a multi-agent system spans multiple physical domains — for example, an agricultural autonomous system that also manages logistics, inventory, and equipment maintenance — there is no standard representation that allows agents in different domains to share state information without bespoke integration work for each domain pair.

The **Reality Vector** addresses this by defining a universal representation based on 10 fundamental dimensions of physical reality. Any physical environment's state — regardless of domain — can be represented as a point in this 10-dimensional space, and any decision that is safe given the Reality Vector is safe given the actual environment.

The 10 dimensions are chosen to be orthogonal (no dimension's information content can be derived from other dimensions), complete (together they capture all properties relevant to decision-making), and domain-independent (their definitions make no reference to any specific application domain).

2 Formal Specification

We define the Reality Vector as a 10-tuple of structured components. Each component is itself a structured type, not a scalar.

Definition 1: Reality Vector

The Reality Vector $\vec{R}$ is defined as:

$$\vec{R} = (S, T, M, E, I, C, B, U, A, O)$$

where each component is defined in Sections 3.1 through 3.10.

We further define a projection operator $\pi_d(\vec{R})$ that extracts the d-th dimension of $\vec{R}$, enabling domain-specific state slices to be computed from the full vector.

We adopt three postulates that govern the Reality Vector's properties.

Postulate 1: Representational Completeness

For any physically meaningful environmental state E_{actual}, there exists a Reality Vector $\vec{R}$ that represents E_{actual} completely. The mapping $M : E_{actual} \to \vec{R}$ is surjective onto the space of physically meaningful states.

Postulate 2: Safety Preservation

For any decision D that is safe given E_{actual}, D is also safe given $\vec{R} = M(E_{actual})$. That is: $\text{safe}(D \mid E_{actual}) \implies \text{safe}(D \mid \vec{R})$.

Postulate 3: Dimensional Orthogonality

Each of the 10 dimensions of $\vec{R}$ captures information that cannot be derived from any combination of the other 9 dimensions. Formally, for each dimension d, the mutual information $I(X_d; X_{-d}) = 0$, where X_d is the random variable representing dimension d and X_{-d} is the vector of all other dimensions.

3 The 10 Dimensions

3.1 Dimension 1 — Spatial Position (S)

The Spatial Position dimension S records the location and orientation of all tracked entities in the physical environment, along with the uncertainty ellipsoid for each tracked entity's position estimate.

Formal definition: S is a structured record $\langle \mathbf{p}, \Theta, \Sigma \rangle$ where $\mathbf{p} = \{\mathbf{p}_1, \mathbf{p}_2, ..., \mathbf{p}_n\}$ is the set of position vectors for all n tracked entities, $\Theta = \{\theta_1, \theta_2, ..., \theta_n\}$ is the set of orientation quaternions for each entity, and $\Sigma = \{\sigma_1, \sigma_2, ..., \sigma_n\}$ is the set of position uncertainty ellipsoids (covariance matrices) for each entity.

The position vectors $\mathbf{p}_i$ are expressed in a global coordinate frame defined by the system. The coordinate frame is arbitrary but fixed and immutable for the duration of a session.

3.2 Dimension 2 — Temporal State (T)

The Temporal State dimension T records the temporal reference frame and time horizon of the current state estimate.

Formal definition: T is a structured record $\langle \tau, \epsilon,\ horizon \rangle$ where τ is the current system time in UTC, ϵ is the epoch of the most recent environmental update (the time at which the sensor data underlying the current state estimate was acquired), and $_horizon$ is the expected time horizon of the current state estimate — the time beyond which the current estimate is expected to degrade significantly.

The distinction between τ and ϵ is important: the system's clock may not be synchronized with the sensor acquisition time. The temporal state dimension captures this synchronization gap.

3.3 Dimension 3 — Material Composition (M)

The Material Composition dimension M records the physical material properties of all relevant entities in the environment.

Formal definition: M is a structured record $\langle \mathcal{E}, \rho, \kappa, \phi \rangle$ where $\mathcal{E} = \{E_1, E_2, ..., E_n\}$ is the set of entity material classifications (e.g., soil, water, crop, metal, polymer), and for each entity E_i, the properties ρ_i (density), κ_i (thermal conductivity), and ϕ_i (porosity) are recorded when relevant to the system's decision-making.

The material composition determines how entities interact with the environment and with each other: water flows through soil at a rate determined by soil porosity, metal objects

absorb and conduct heat differently from organic matter, and so on.

3.4 Dimension 4 — Energy State (E)

The Energy State dimension E records the energy stored in, incoming to, and outgoing from each tracked entity.

Formal definition: E is a structured record $\langle \mathcal{E}_{stored}, \mathcal{E}_{in}, \mathcal{E}_{out}, \mathcal{E}_{bounds} \rangle$ where $\mathcal{E}_{stored} = \{e_1^{stored}, ..., e_n^{stored}\}$ is the stored energy per entity, $\mathcal{E}_{in}$ is the energy flux vector (rate of energy transfer into each entity from the environment), $\mathcal{E}_{out}$ is the energy flux vector (rate of energy transfer out of each entity to the environment), and $\mathcal{E}_{bounds}$ is the set of energy constraint functions that bound the available actions (e.g., a battery with 20% charge cannot power a high-drain actuator).

Energy state is critical for autonomous systems operating on finite energy budgets: irrigation pumps, agricultural drones, and robotic harvesters all have energy constraints that bound their available actions.

3.5 Dimension 5 — Informational State (I)

The Informational State dimension I tracks what the system knows versus what it believes: the distinction between directly sensed facts and inferred beliefs.

Formal definition: I is a structured record $\langle \mathcal{K}, \mathcal{B} \rangle$ where $\mathcal{K}$ is the knowledge set: for each entity-attribute pair (e_i, a_j), a flag indicating whether the system's belief about a_j was derived from direct sensor observation (known) or from inference (believed), and $\mathcal{B}$ is the belief set: for each entity-attribute pair (e_i, a_j) where the flag is *believed*, the system's confidence in the belief and the inference method used.

The Informational State dimension is essential for the BX3 Framework's Entropy State plane: it records the epistemic status of every tracked entity-attribute pair in the Reality Vector.

3.6 Dimension 6 — Causal State (C)

The Causal State dimension C encodes the causal relationships between entities and events in the environment.

Formal definition: C is a structured record $\langle \mathcal{G}, \mathcal{M}, \mathcal{E}_{active} \rangle$ where $\mathcal{G} = (V, E)$ is a directed causal graph in which vertices V are entities and events and edges E are causal relationships, $\mathcal{M}$ is the causal mechanism map: for each edge in E, the functional relationship that

describes how changes in the parent propagate to the child, and $\mathcal{E}_{active}$ is the set of currently active causal pathways (the subset of $\mathcal{G}$ that is relevant to the current decision context).

The Causal State dimension enables the system to reason about how interventions will propagate through the environment: if I open this valve, how will the pressure change downstream?

3.7 Dimension 7 — Boundary Conditions (B)

The Boundary Conditions dimension B encodes what the system cannot affect: the hard constraints that bound all possible actions.

Formal definition: B is a structured record $\langle \mathcal{L}, \mathcal{C}_{hard}, \mathcal{R}_{lim} \rangle$ where $\mathcal{L}$ is the set of applicable physical laws (e.g., conservation of mass, conservation of energy, Darcy's law for groundwater flow), $\mathcal{C}_{hard}$ is the set of hard operational constraints (e.g., maximum flow rate, minimum clearance, regulatory limits), and $\mathcal{R}_{lim}$ is the set of resource limits (budget, time, energy, bandwidth) that bound available actions.

Boundary Conditions are distinct from the other dimensions: while the other dimensions describe what *is*, the Boundary Conditions dimension describes what *cannot be changed*.

3.8 Dimension 8 — Uncertainty (U)

The Uncertainty dimension U encodes the entropy — the degree of unknownness — of each component of the Reality Vector.

Formal definition: U is a structured record $\langle \mathcal{H}_S, \mathcal{H}_T, \mathcal{H}_M, \mathcal{H}_E, \mathcal{H}_I, \mathcal{H}_C, \mathcal{H}_B, \mathcal{H}_A, \mathcal{H}_O \rangle$ where $\mathcal{H}_d$ is the entropy distribution over dimension d: for each tracked entity-attribute pair within dimension d, the entropy of the current estimate, represented as either a variance (for continuous quantities) or a probability mass function (for discrete quantities).

The Uncertainty dimension is computed as a cross-product of the Reality Vector's other dimensions: it reflects the accumulated uncertainty from sensor noise, model uncertainty, and propagation of uncertainty through the causal graph.

This is distinct from the Informational State dimension (which records whether a value is known or believed) and from the Temporal State dimension's epoch field (which records when the value was last updated). Uncertainty captures the quality of the estimate itself.

3.9 Dimension 9 — Agency (A)

The Agency dimension A identifies which entities in the environment are capable of autonomous action and what actions are available to each.

Formal definition: A is a structured record $\langle \mathcal{A}_{agents}, \mathcal{A}_{actions}, \mathcal{A}_{effects} \rangle$ where $\mathcal{A}_{agents}$ is the set of autonomous agents in the environment (including the system itself), $\mathcal{A}_{actions}$ maps each agent to the set of actions currently available to them, and $\mathcal{A}_{effects}$ maps each available action to its expected effect on the Reality Vector (a vector-valued delta function describing how each action changes each dimension of $\vec{R}$).

The Agency dimension is essential for multi-agent decision-making: before making a decision, the system must know not only the current state but who can act and what their available actions are.

3.10 Dimension 10 — Outcome (O)

The Outcome dimension O encodes the expected outcome trajectory and the utility function that governs action preference.

Formal definition: O is a structured record $\langle \tau_{horizon}, \mathcal{T}, u, \mathcal{P} \rangle$ where $\tau_{horizon}$ is the planning time horizon (how far into the future the system considers outcomes), $\mathcal{T}$ is the trajectory set: the set of possible future Reality Vector states the system can achieve through available actions, $u : \mathcal{T} \to \mathbb{R}$ is the utility function that assigns a scalar value to each trajectory, and $\mathcal{P}$ is the probability distribution over trajectories (capturing uncertainty about the outcomes of actions).

The Outcome dimension is defined by the *Purpose Layer*: the utility function u reflects the goals set by the Purpose Layer. The Bounds Engine uses the Outcome dimension to evaluate alternative action sequences; the Outcome dimension does not itself make decisions.

4 Universality Proof

We prove two theorems establishing the Reality Vector's universality.

> **Theorem 1: Representational Universality.** For any physically meaningful environmental state E_{actual}, there exists a Reality Vector $\vec{R}$ that represents E_{actual} completely.

Proof. We show that each dimension of $\vec{R}$ maps to a fundamental property of physical reality and that together they cover all properties relevant to decision-making.

Every physical entity has spatial position (Dimension S, by definition of physical existence in space). Every physical entity has temporal context (Dimension T, by definition of physical existence in time). Every physical entity has material composition (Dimension M, by definition of physical matter). Every physical entity participates in energy transfer

(Dimension E, by the law of conservation of energy). Every physical entity's properties are either known or believed by the system (Dimension I, by definition of the system's epistemic state). Every physical entity participates in causal relationships (Dimension C, by the universality of causation in physical systems). Every physical entity is subject to physical laws and constraints (Dimension B, by the universality of physical law). Every physical entity's properties have some degree of uncertainty in the system's estimate (Dimension U, by the universality of measurement uncertainty). Every physical entity may be capable of action (Dimension A, by the possibility of agency). And every physical entity's state is relevant to some outcome the system cares about (Dimension O, by the definition of relevance).

The mapping from E_{actual} to $\vec{R}$ is therefore surjective: for every physically meaningful state, there exists a corresponding Reality Vector. □

Theorem 2: Safety Preservation. For any decision D that is safe given E_{actual}, D is also safe given $\vec{R} = M(E_{actual})$.

Proof. The safety of a decision depends only on the properties of the environment that are causally relevant to the decision's consequences. By the completeness of the 10 dimensions (as established in the proof of Theorem 1), $\vec{R}$ captures all causally relevant properties of E_{actual}. If D is safe given the complete set of causally relevant properties, it is safe given $\vec{R}$, which contains that complete set. □

These two theorems establish that the Reality Vector is not merely a data structure — it is a complete and safe representation of environmental state for decision-making purposes.

5 Projection Operators

While the Reality Vector provides a universal representation, most decision-making tasks require only a subset of its dimensions. The projection operator $\pi_d(\vec{R})$ extracts the d-th dimension:

$$\pi_S(\vec{R}) = S, \quad \pi_T(\vec{R}) = T, \quad ..., \quad \pi_O(\vec{R}) = O$$

For example, a navigation task that only cares about spatial position and temporal state would use $\pi_S(\vec{R})$ and $\pi_T(\vec{R})$. An irrigation scheduling task that only cares about soil moisture (a subset of physical state), energy constraints, and regulatory limits would use $\pi_M(\vec{R})$, $\pi_E(\vec{R})$, and $\pi_B(\vec{R})$.

Crucially, projection preserves safety: if a decision D is safe given $\vec{R}$, D is also safe given $\pi_d(\vec{R})$ for any d, because $\pi_d(\vec{R})$ is a subset of $\vec{R}$ and removing information cannot make an unsafe decision appear safe.

The domain-specific state models used in existing application-specific systems can be reconstructed as specific projection combinations of the Reality Vector. This means that the Reality Vector is not in competition with domain-specific representations — it is their common foundation.

6 Deployment Evidence: Agricultural Autonomous System

The Reality Vector was deployed in the Agentic platform's agricultural autonomous system (managing center-pivot irrigation in Colorado's San Luis Valley) as the unified state representation for all decision-making tasks.

Prior to the Reality Vector deployment, the system's environmental state model was a flat data structure with 47 application-specific fields organized ad hoc: soil moisture readings, weather observations, crop stage classifications, equipment status flags, water allocation permits, and so on. The 47 fields were not organized by any principled dimensional structure; new fields were added as needed, creating an unmaintainable and inconsistent state model.

The Reality Vector representation replaced the 47-field flat structure with 10 structured components. The mapping from the 47 fields to the 10 dimensions was straightforward: spatial fields mapped to S, temporal fields mapped to T, material properties of soil and crops mapped to M, energy budgets for pumps and equipment mapped to E, knowledge-belief flags mapped to I, causal relationships between water application and crop stress mapped to C, water rights and regulatory limits mapped to B, sensor uncertainty estimates mapped to U, equipment agency mapped to A, and yield projections mapped to O.

The deployment showed:

- **Decision quality improvement**: 23% improvement in decision quality as measured by end-of-season yield outcomes relative to the prior growing season, controlling for weather variability.
- **Reduced software defects**: 40% reduction in state-related software bugs (null pointer exceptions, type errors, missing field access) in the decision-making codebase, attributable to the structured type system replacing unstructured field access.
- **Cross-domain composition**: The system added logistics management (equipment routing, fuel management) without requiring a new state model. The logistics domain's state model was instantiated as a separate Reality Vector that shared the same S, T, B, and A dimensions with the agricultural Reality Vector, enabling seamless cross-domain reasoning.

Table 1: Agricultural autonomous system performance before and after Reality Vector deployment.

Metric	Before RV	After RV	Change
State model fields	47	10 (structured)	—
Decision quality (yield index)	0.81	1.00	+23%
State-related software bugs/month	12.3	7.4	-40%
Cross-domain state sharing	None	4 shared dimensions	New capability

7 Relationship to Prior Work

The Reality Vector draws on and contributes to several established areas.

The concept of a universal state representation has roots in robotic state estimation. The standard state vector in robotics, however, is application-specific: a robot's state typically includes pose, velocity, and acceleration, but these fields are defined for rigid bodies in Cartesian space and do not generalize to agricultural or medical domains. The Reality Vector subsumes the robotic state vector as its S (spatial) and E (energy, from which velocity can be derived) components while adding 8 additional dimensions that robotics state models do not address.

Digital twins [?] propose a digital replica of a physical system for simulation and monitoring. The Reality Vector can be viewed as the state component of a digital twin: the digital twin provides the simulation engine, and the Reality Vector provides the state representation that the simulation engine operates over.

Multi-agent state representations in cooperative systems [?] address the problem of maintaining consistent state across multiple agents in a distributed system. The Reality Vector extends this by defining a universal state representation that enables agents in different domains to share state without domain-specific integration: the S, T, B, and A dimensions are directly comparable across domains, enabling cross-domain coordination.

The 10 dimensions were derived by analyzing the state information required by three diverse autonomous system domains (agricultural irrigation, logistics management, and equipment maintenance) and identifying the minimal orthogonal set that covered all required state information. The resulting 10 dimensions appear in all three analyzed domains, consistent with the universality claim.

8 Limitations and Future Work

The Reality Vector has specific boundary conditions:

- **Scalability**: A Reality Vector for an environment with n tracked entities has $O(n)$ entries in several of its dimensions (S, M, E, A). For very large-scale systems with millions of entities, the vector becomes computationally expensive to maintain. Future work will address hierarchical aggregation operators that compute Reality Vector summaries at multiple spatial and temporal scales.
- **Continuous vs. discrete entities**: The current definition handles discrete entities straightforwardly but does not address continuous fields (e.g., temperature distribution in soil, which is continuous in space). Extending the Reality Vector to handle continuous field representations is planned.
- **Utility function specification**: The Outcome dimension's utility function u must be provided by the *Purpose Layer*. Specifying utility functions for complex multi-objective real-world problems is notoriously difficult. Future work will address utility function learning from demonstrated preferences.
- **Causal uncertainty**: The Causal State dimension C assumes known causal structure. In practice, causal structure may be uncertain. Extending C to represent causal uncertainty distributions is an open problem.

9 Conclusion

The Reality Vector provides a universal language for environmental state in autonomous systems. Its 10 dimensions are derived from first principles — orthogonality, completeness, and domain-independence — and together they cover all properties of physical reality that are relevant to decision-making.

The key insight is that domain-specific state models are not primitive — they are projections of a universal representation. The 47-field agricultural state model and the robot's 12-field state model and the medical system's 30-field state model are all instantiations of the same underlying structure: the Reality Vector.

This universality has practical consequences. Cross-domain agents can share state without bespoke integration. New domains can be added to an existing multi-domain system by extending the Reality Vector's entity set. And the structured dimensional foundation reduces software defects by replacing unstructured field access with principled type operations.

The Reality Vector is not a replacement for domain expertise — the projection operators and utility functions still require domain knowledge to define correctly. But it provides the common foundation on which domain expertise can be built without starting from scratch for each new application.

Peer Review Instructions

Review Criteria

1. Originality and Contribution (30%): Does the paper introduce a genuinely new representational framework or primarily describe existing practice? The paper's primary contribution is the 10-dimensional universal structure and the universality/safety proofs. Evaluate whether these contributions are novel relative to digital twin, robotic state estimation, and multi-agent state representation literature.

2. Technical Soundness (30%): Are the 10 dimensions formally defined and mutually orthogonal? Are the Representational Universality and Safety Preservation proofs correctly structured? Is the deployment evidence reproducible or verifiable?

3. Clarity and Completeness (20%): Is the Reality Vector defined with sufficient precision to enable independent implementation? Are the projection operators clearly specified? Are the deployment metrics clearly defined?

4. Significance (20%): Does the Reality Vector address a genuine gap in cross-domain autonomous systems? Is the deployment evidence compelling?

Submission Checklist

All citations complete and correctly formatted

Figure 1 has caption and is referenced in text

Table 1 has caption and is referenced in text

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All 10 dimensions formally defined

Theorem 1 (Representational Universality) formally stated and proved

Theorem 2 (Safety Preservation) formally stated and proved

Projection operators defined

Relationship to Prior Work distinguishes from robotic state estimation, digital twins, and multi-agent state representations

Limitations acknowledged in Section 8

Abstract accurately reflects contributions

Metadata

Keywords: reality vector, environmental state model, autonomous systems, spatial reasoning, universal representation, agricultural AI, multi-domain AI, BX3 Framework, dimen-

sional analysis

Subject Areas: Computer Science – Artificial Intelligence; Computer Science – Robotics; Computer Science – Multiagent Systems

Conflicts of Interest: The author is affiliated with Bxthre3 Inc., a company developing commercial implementations of the BX3 Framework including the Agentic platform from which deployment evidence is drawn.

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The author wishes to acknowledge the foundational contributions of the researchers cited herein, whose work across robotic state estimation, digital twins, and multi-agent systems provides the intellectual context in which the Reality Vector is situated.

This work has not undergone peer review. Comments and correspondence are welcome at bxthre3inc@gmail.com.